

THE AI & DELIBERATION REALM

Untrusted Compute Silicon (MPU / GPU / NPU)

Core Architectural Focus:

- High-capacity open-world generalizability (7B–70B VLMs).
- Multi-modal attention & dense spatial patch tokens.
- Iterative diffusion trajectory denoising ($H = 16\text{--}64$ steps).

Execution Characteristics:

- Variable compute latency (100–1000 ms, long P_{99} tails).
- Non-zero statistical hallucination & stochastic drift.
- System Role: **Untrusted Proposal Generator**.



THE PHYSICAL & DYNAMICS REALM

Deterministic Real-Time Microcontroller (MCU)

Core Architectural Focus:

- Unyielding Newtonian physics ($M\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{g} = \boldsymbol{\tau}$).
- Kinetic momentum ($p = mv$) & stopping distance (d_{stop}).
- Inverter electromagnetic torque ($K_t i_q$) & coil rise (L/R).

Execution Characteristics:

- Microsecond determinism ($1000\text{ Hz} \pm 5\text{ }\mu\text{s}$, 'malloc = 0').
- Zero-tolerance for delayed or missing actuation updates.
- System Role: **Sole Execution Authority (Permission Gate)**.

THE THREE ARCHITECTURAL CONTRACTS RECONCILING SCALE AND DYNAMICS

1. **Proposal-Permission Split:** MPU proposes candidate trajectories $\rightarrow$ MCU audits with Control Barrier Functions.
2. **Action Chunking:** Generative models predict H -step horizons, amortizing non-zero neural compute delays.
3. **Expiring Intent Leases:** Hardware watchdog timers enforce monotonic timeouts, guaranteeing safe fallback on latency spikes.